

WASP-10b

R-0678

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_241109 · created 2026-07-27T19:16:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460623.7238 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1403 +/- 0.0094
Transit depth (Rp/Rs)^2	1.97 +/- 0.26 [%]
Orbital Inclination (inc)	89.62 +/- 0.55
Airmass coefficient 1 (a1)	1.01 +/- 0.017
Airmass coefficient 2 (a2)	-0.0069 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	1.68 %
Best Comparison Star	#2 - [620, 296]
Optimal Aperture	3.52
Optimal Annulus	3.83
Transit Duration (day)	0.093 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1403 ± 0.0094 vs 0.1592 (deviation 1.21σ)
Duration fit vs published	0.093 d vs 0.0946 d (deviation 0.86σ)
Mid-time O–C	-1.5 min
Depth SNR	8.9
Residual scatter	1.68 %

Light curve

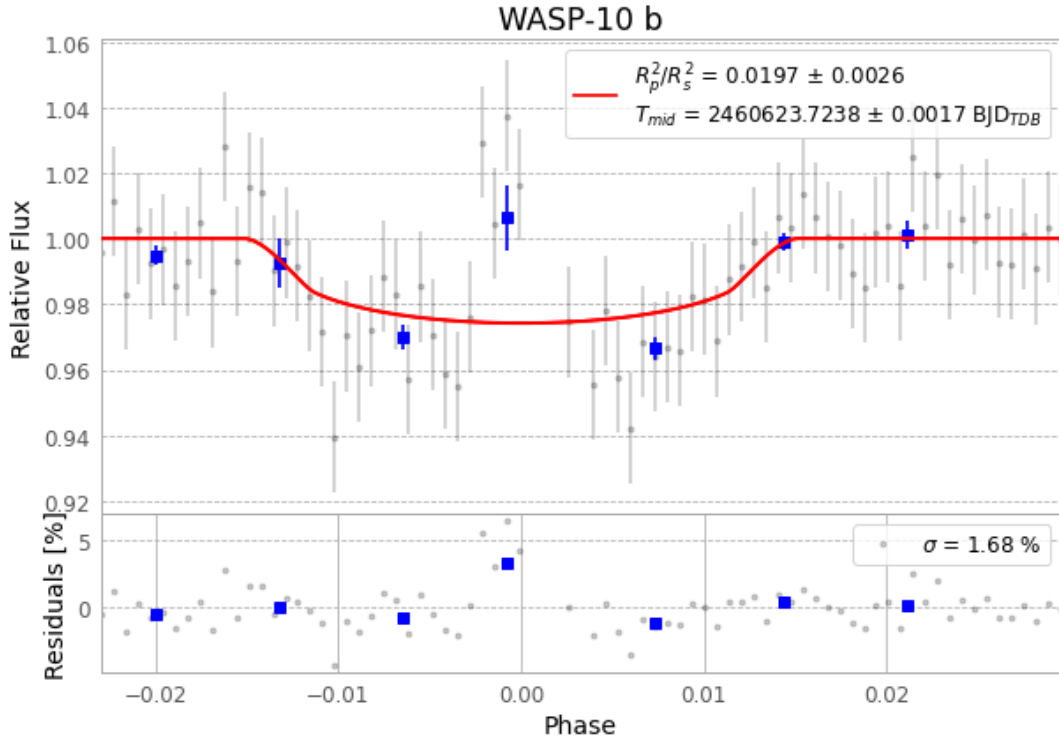


Figure 1 — FinalLightCurve_WASP-10 b_2024-11-08.png (SHA-256 fe7a0c2078554e1a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 289], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 69c5275ea1292fa5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

79 FITS frames (52 MB), WASP-10241109033315.FITS → WASP-10241109072715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-241109.log (c4d232a473c8...), FinalLightCurve_WASP-10 b_2024-11-08.png (fe7a0c207855...), FinalLightCurve_WASP-10 b_2024-11-08.csv (b6a178d62574...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:16Z.